How school type and socioeconomic status shapes average SAT scores in Chicago (SY 2024-2025)

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Background and Importance

Growing up in the city of Chicago, many 8th grade students are often met with a choice of either going to their neighborhood high school or trying to test into a selective enrollment high school. Selective enrollment schools are said to have better education quality and be more challenging for students as well. I was interested in seeing if there was a difference in SAT scores between neighborhood schools and selective enrollment given the fact that there has been a long standing ideal that selective enrollment schools provide better education and prepare students for college. I think it is also important to note that education quality may also vary depending on school location, so I wanted to also compare to the median incomes by zip code. Do these factors truly play a difference in the education of children in Chicago?

Specific Research Questions

There are multiple research questions that I was hoping to address when doing my research including:

- How do average SAT scores differ between neighborhood and selective enrollment high schools in Chicago?
- Does average SAT score differ between zip code, does a higher median income zip code have higher SAT score and vice versa?

Data Collection Methodology

The SAT data was found through the Chicago Data Portal. It was school progress reports using data from 2023-24. It contained all the CPS schools and information such as school name, zip, SAT scores, school type and more. Each school collects data to be sent to the network to analyze.

Median Income data was found on simple maps. It used data from ACS 2023. I had to transfer the data into excel and export as an CSV file to R studio.

Variables Used

The variables I used were:

1.SAT score: A continuous variable. I was able to remove any values that were equal to 0, grouped it by school type, and also by zip code. Purpose was to be able to compare the SAT scores by zip code and school type.

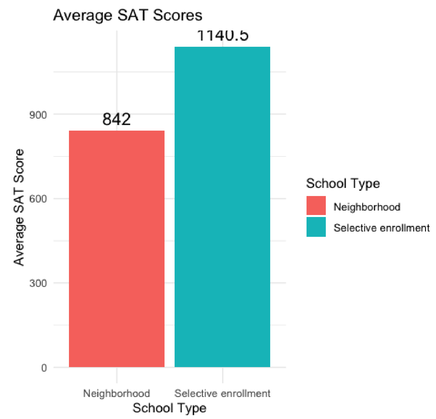
2.School Type: A categorical variable. I was able to filter out to only keep neighborhood and selective enrollment in original data set. Purpose was to be able to see if there was a difference in SAT scores between the 2.

3. Zip Code: Numeric converted to Character to be able to be joined with income data and also data to create map. Purpose was to see if there was a difference in SAT score base on location.

4.Median Income: Continuous variable. Was able to join with SAT score dataset using Zip code. Purpose was to see the difference of income throughout the city of Chicago.

Analysis Approach & Results

To analyze the data, I decided to create a bar graph, 2 maps (one showing the different SAT scores by Zip and another median household income by zip), and 2 tables that showed compared SAT scores between the zip codes and school types.



This bar graph shows average SAT scores from 2023-2024 school year. It compares neighborhood vs. selective enrollment. Selective enrollment has a higher average SAT.

Highest and Lowest Median Household Incomes and their average SAT A.

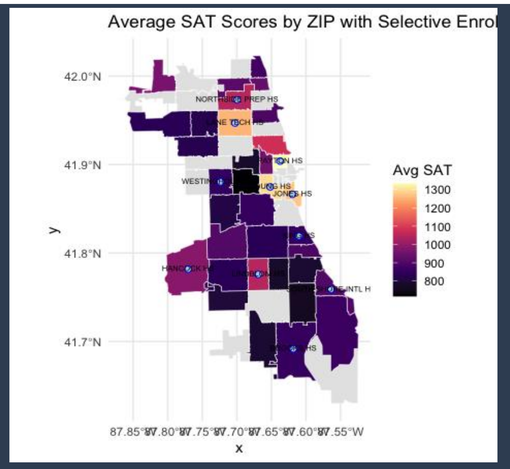
Zip Code	Median Household Income	Average SAT
60624 Garfield Park	$32,607	849
60621 Englewood	$33,235	778
60614 Lincoln Park	$139,561	1081
60642 Near North & West Town	$141,179	952

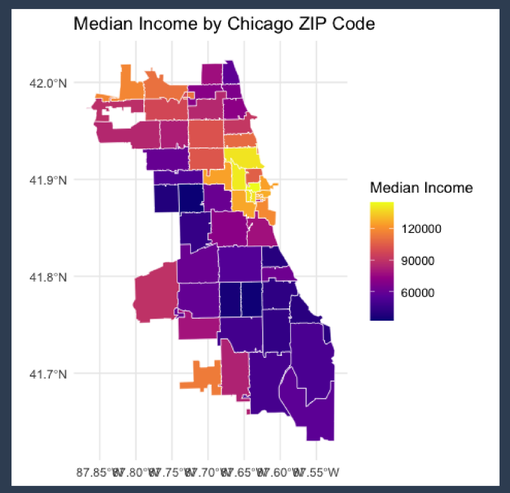
* 2 lowest & 2 highest median income zip codes

Higher Income zip code areas have a higher average SAT compared to lower income. Selective Enrollment had higher Mean, Median, Min and Max compared to neighborhood.

B.

School Type	Number of Schools	Mean SAT	Median SAT	Min SAT	Max SAT
Neighborhood	42	842.00	813	716	1081
Selective enrollment	11	1140.55	1111	901	1334





Conclusions

When considering table A , it can be noted that the higher income zip codes did have higher average SAT scores compared to the lower income. There is some difference when comparing the higher income zip codes together. Zip code 60614 had a higher SAT average than 60642 even though it had a lower median household income. Same for the lower income zip codes. Increases in income does not always correspond to higher SAT average. There can be some correlation between income and SAT scores but there can be other factors that influence such as school type, school demographics, and resources. When looking at the maps, there is a lot of variance in both SAT scores and median income between the zip codes. The higher average SAT areas did have selective enrollment schools and for the most part had higher median income as seen by the lighter peach/yellow tones.

When considering the bar graph and table B, it can be noted that the selective enrollment schools had higher SAT scores all around. There can be some correlation but there is a variance in education depending on the selective enrollment school.

Overall, higher income areas and selective enrollment schools tend to show higher SAT performance but do not solely determine it since there is shown to be a variance looking at zip codes.

Limitations and possible future directions

The limitations I would like to address is that with using American Community Survey data there may be some slight variance in median income since it is using income data from 2023. I would believe it would only make a slight difference not very noticeable from when the SAT scores were being reported. While the school year is for 2024-2025, the CPS data being used is from 2023-2024. In the future, I would like to be able to combine the data into one map to show the median income but also plot points geographically where schools are located but be able to distinguish between neighborhood vs selective enrollment and at the same time have the points vary in size depending on the average SAT score. It would also be worth to investigate other variables and how much they play a role in SAT scores. It would also be worth to take a look at other school types as well.

References

Median household income:
https://simplemaps.com/city/chicago/zips/income-household-median

CPS SAT Data:
https://data.cityofchicago.org/Education/Chicago-Public-Schools-School-Progress-Reports-SY2/twrw-chuq/about_data